Experiment 1: Setup a Web Application Project Using Express.js and Run the Server

Aim: To setup an Express.js web application and test its working.

Procedure:
1. Create project folder:
   mkdir express-app
   cd express-app
2. Initialize project:
   npm init -y / npm init -y (creates package.json file)
3. Install Express:
   npm install express
4. Create app.js manually inside express-app.
5. Add code:

```
const express=require('express');
const app=express();
app.get('/',(req,res)=>{res.send('Hello, Express.js!');});
app.listen(3000,()=>console.log('Server running at http://localhost:3000'));
```

6. Run application:
   node app.js [$npm start]
7. Open http://localhost:3000

Expected Output:
Browser: Hello, Express.js!
Terminal: Server running at http://localhost:3000

Experiment 2: Middleware, Controllers and Routers

Aim: To demonstrate middleware, controllers and routers in Express.js.

Procedure/steps:
1. Create project:
   mkdir middleware-demo
   cd middleware-demo
   npm init -y
   npm install express

2. Create folders:

(In solution explorer: right click on project name and select 'Add' and click 'New folder')

a. controllers
b. middleware
c. routes (already exist)

3. Create files:

controllers/userController.js
middleware/logger.js
routes/userRoutes.js (Note: we already have user.js and use it accordingly)
app.js

4. logger.js:

```
const logger=(req,res,next)=>{console.log(req.method);console.log(req.url);next();};
module.exports=logger;
```

5. userController.js:

```
exports.getUser=(req,res)=>{res.send('User Controller Executed');};
```

6. userRoutes.js/user.js:

```
const express=require('express');
const router=express.Router();
const userController=require('../controllers/userController');
router.get('/',userController.getUser);
module.exports=router;
```

7. app.js:

```
const express=require('express');
const app=express();
const logger=require('./middleware/logger');
const userRoutes=require('./routes/userRoutes');
app.use(logger);
app.use('/users',userRoutes);
app.listen(3000,()=>console.log('Server running at http://localhost:3000'));
```

8. Run:

node app.js

9. Open http://localhost:3000/users

Expected Output:
Browser: User Controller Executed
Terminal:
Request Method: GET
Request URL: /users